

Bipartite negation in Nduindui: empirical generalisations and theoretical challenges

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Abstract

The goal of this paper is to explore negation in Nduindui, drawing out empirical generalisations regarding negation and the implications of such generalisations in terms of analysing the clause structure in the language. Nduindui shares an interesting feature with many of its neighbours: consistently exhibiting discontinuous bipartite negation, where the two negators *he* and *tea* encircle verbal material. This paper presents novel data regarding negation in Nduindui and its interaction with other elements in the clause like tense and aspect. Empirical generalisations are formed from the attested patterns of negation, and a comparative discussion is provided with other languages in the region. The paper then explores what might be the best way to capture the generalisations within a generative syntactic framework. One of the most widely accepted theoretical analyses for bipartite negation is based on French (see Pollock, 1989), where the two negators are argued to be the head and the specifier of the same phrase/NegP; this paper discusses how the Nduindui patterns present challenges to such an analysis. The paper then evaluates the consequences of extending three alternative analyses that can capture bipartite negation: (a) the PolP account of Oosthuizen (1998) and also Biberauer (2007, 2008), (b) the Expanded/Articulated NegP of Bell (2004), and (c) an account with two separate NegPs combined with VP/V-movement. The paper demonstrates how the Nduindui patterns remain a challenge to all the existing theoretical approaches, as none of them can explain the patterns without either making wrong predictions or needing additional stipulations.

Keywords: Nduindui, Ambae, syntax, negation, bipartite negation, Vanuatu

1. Introduction

Nduindui (ISO: ndd) is the exonym for a language spoken on the island of Ambae, Vanuatu (Ambae is also referred to as Aoba and Omba, and as Leper's Island by older sources), of the Northern Vanuatuan linkage (Lynch, Ross, & Crowley, 2002).¹ Nduindui is the place name

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¹ The data presented in this paper was collected in 2021 at the University of Auckland Waipapa Taumata Rau, in Auckland New Zealand. The data for was collected through structured elicitation sessions with a native speaker, following standard practices in generative syntax. Elicitation was employed in order to obtain grammaticality judgments on constructed sentences as well as to explore speakers' intuitions about structural contrasts. Participants were asked to evaluate sentences presented in the target language, indicating whether each sentence was acceptable, marginal, or unacceptable. In addition, controlled prompts were used to guide speakers in producing particular constructions of interest. This method allows for systematic testing of theoretical hypotheses by controlling for variation in context and ensuring comparability across examples. Follow-up sessions were used to confirm consistency of judgments. This approach ensures that the data reflect native-speaker intuitions, which

referring to a particular Western Ambae district. Description of the language situation on Ambae has remained relatively sparse and is described differently within different literature. The island is commonly split into two broad language ‘regions’, where the lower west side is labelled as the West-Ambae language and the North-East section is referred to as the Ambae language (see Tryon, 1976; Hyslop, 2001 & François et al., 2015). Nduindui is often understood as a dialect of the broader West-Ambae language, alongside the dialect spoken in the Walaha district (for a more detailed description of the area, see Hyslop, 2001 and Aru, 2015). The languages of Ambae have received relatively little analysis. Aru (2015) provides a basic grammar of Nduindui (which she represents orthographically as ‘Duidui’), based mostly on primary data gathered from native speakers on Ambae, as well as speakers who have migrated to other areas of Vanuatu. Hyslop’s (2001) description of Lolovoli, a dialect spoken in the North-East Ambae region, also provides a more detailed breakdown of other dialects, including some comparison between Lolovoli and Nduindui.²

The goal of this paper is to explore negation in Nduindui, drawing out empirical generalisations regarding negation and the clause structure in the language. Within the wide range of Vanuatuan languages, negation is a topic that has received a great deal of attention. Nduindui shares an interesting feature with many of its neighbours: exhibiting (obligatory) discontinuous bipartite negation, where the two negators *he* and *tea* encircle verbal material. This paper presents data regarding negation in Nduindui and its interaction with other elements. Empirical generalisations are formed from the attested patterns of negation, and comparison is made between potential analyses. One of the most widely accepted theoretical analyses for bipartite negation is based on French (see Pollock, 1989), where the two negators are argued to be part of the same phrase NegP; this paper discusses how the Nduindui patterns present challenges to such an analysis. We then compare accounts in which each of the two negators is a head of a functional projection. In the first, one of the negators, *he*, is the head of NegP, which is present between TP and vP. The other negator *tea* is the head of PolP (Polarity Phrase) which is located higher in the clause structure, in the left periphery. The surface order of elements can be derived by movement of CP to SpecPolP, a movement which is triggered by the EPP feature present in the PolP (similar to analyses proposed for Afrikaans negative concord in Oosthuizen 1998 and Biberauer 2007, 2008). In the second potential account, we discuss the possibility of extending Bell’s (2004) expanded NegP. We then evaluate the possibility of utilising VP/V-raising within a clause structure containing two separate NegPs; with NEG1 *he* the head of the higher NegP and NEG2 *tea* the head of the lower NegP. We use basic X-bar architecture for the theoretical discussion, and explicit formal Minimalist terminology is for the most part avoided outside of Section 4, in the interests of displaying the initial data and generalisations for readers from a wider range of theoretical backgrounds.

The paper is structured as follows. In Section 2, the empirical patterns regarding negation and other elements of the clause are presented. Section 3 briefly describes the negation patterns of several other Vanuatu languages which employ discontinuous negation strategies. Section 4 discusses the possible structural analyses for negation in the language. It is argued in Section 4.1 that an analysis of bipartite negation along the lines of Pollock (1989) for French,

are central to the generative framework, while also providing replicable evidence for syntactic analysis. Special thanks must go to our consultant for sharing their language with us. We are grateful for their contributions and patience; all errors are our own.

² Hyslop (2001) concludes that Nduindui is a “distinct” language from North-East-Ambae (of which Lolovoli is a dialect) (Hyslop, 2001, p. 2 – 3). Aru (2015) distinguishes between the Walaha and Nduindui dialects, but prefers the name ‘West Ambae’ for the broader language. The status of the language varieties spoken on Ambae and their relationships to one another remains unclear.

cannot be extended to capture the Nduindui facts, and the implications of three alternate proposals designed to capture the negation patterns attested in Nduindui are detailed in Sections 4.2., 4.3. and 4.4. Section 5 concludes the paper.

2. Empirical patterns: Standard Negation

Following Miestamo's (2005) definition of 'standard negation' to mean a language's basic or default strategy through which clausal negation of a declarative main verb is achieved, Nduindui's standard negation strategy involves two discontinuous negative markers, both of which are obligatory. Nduindui is an SVO language, with tense and aspect marking appearing most often pre-verbally. In (1) and (2) the negative markers encircle just an intransitive verb. In contrast, in (3), the negative markers can be seen to encircle the verb and the object. In other words, NEG2, *tea*, appears post-object.

- | | | | | | | | |
|-----|--|------------|------|---------|------------------|-------|------|
| (1) | a-ia | u | he | maturu | tea ³ | | |
| | NOM-3SG | 3SG.SUBJ.R | NEG1 | sleep | NEG2 | | |
| | 'He is not sleeping.' | | | | | | |
| (2) | a | taŋaloi | he | kankani | tea | | |
| | NOM | man | NEG1 | RED.eat | NEG2 | | |
| | 'The man is not eating/The man did not eat.' | | | | | | |
| (3) | a | taŋaloi | he | kani | na | mango | tea |
| | NOM | man | NEG1 | eat | ACC | mango | NEG2 |
| | 'The man didn't eat/isn't eating the mango.' | | | | | | |

Under the cross-linguistic assumption that the verb and object together form a constituent, standard negation in Nduindui thus appears to encircle this entire phrase: the VP. Both negators, pre-verbal NEG1 *he* and post-object NEG2 *tea*, are obligatory. The omission of either one is judged ungrammatical by speakers. This is illustrated below, where the negated utterance in (5) is judged as ungrammatical if either NEG2 *tea* or NEG1 *he* is omitted (in 6 and 7 respectively). If only the verb, rather than the entire verb phrase, appears between NEG1 and NEG2, the utterance is also ungrammatical, as attempted in (8), where the object appears to the right of NEG2.

- | | | | | | | |
|-----|---------------------------------|------|-----|-------|-------|------|
| (4) | John | tai | na | kambu | | |
| | John | cut | ACC | fire | | |
| | 'John cut the firewood.' | | | | | |
| (5) | John | he | tai | na | kambu | tea |
| | John | NEG1 | cut | ACC | fire | NEG2 |
| | 'John didn't cut the firewood.' | | | | | |
| (6) | *John | he | tai | na | kambu | |
| | John | NEG1 | cut | ACC | fire | |
| (7) | *John | tai | na | kambu | tea | |
| | John | cut | ACC | fire | NEG2 | |

³ The transcription style within this paper uses a standard Oceanic vowel inventory: [a, e, i, o, u]. One /r/ phoneme has been assumed, which may be produced variably, but has been transcribed consistently as [r] for readability. For the same reason, the language's systems of prenasalisation and labialisation have been transcribed as consonant sequences. Standard IPA transcription of all other consonants has been followed.

- (8) *John he tai tea na kambu
 John NEG1 cut NEG2 ACC fire

However, this pattern is not true with clausal complements. Nduindui embedded clause complements do not appear in the same position as standard direct objects, between the verb and NEG2 *tea*. Instead, NEG2 *tea* intervenes between the verb and its CP complement, as below in (9), where the subordinate clause ‘to eat chicken’ (more literally ‘that I eat chicken’) appears to the right of NEG2.

- (9) na he lengaia tea kwara na kani toa
 1SG NEG1 want NEG2 COMP 1SG eat chicken
 ‘I do not want to eat chicken.’

(Nduindui; adapted from Aru, 2015, p. 165.)

Another issue of relevance in discussing the verbal domain concerns the ordering of adverbs. Aru (2015) provides evidence of adverbial material appearing to the right of NEG2 *tea*, as below in (10), as well as to the left of NEG2 *tea*. She claims that *tea* “can precede the adverb or follow it” (Aru, 2015, p. 158).

- (10) ra he βano tea noinoa
 3PL.R NEG1 go NEG2 yesterday
 ‘They did not go yesterday.’

(Nduindui: data adapted from Aru, 2015, p. 104.)

We are concerned also with the ordering of negation in relation to other clausal elements, namely tense and aspect, in order to shed light on the structural position of the negation. As previous translations provided for (2) and (3) suggest, the interpretation of tense in Nduindui discourse is often reliant on more than overt marking. Structures with no overt tense marking may be interpreted as occurring in the present, or the past. However, some overt tense marking is present in the language. Aspect marking is also present in the progressive/continuous marker: *tuku*. Pre-verbal future tense marking is seen to remain pre-negation – outside the scope of the negative markers encircling the VP, as in (9) and (10). In contrast, the aspect marker in (11) is positioned within the two encircling negative markers, immediately pre-verbal.

- (11) a-ia ne he βaluk na mbuk tea
 NOM-3SG FUT NEG1 read ACC book NEG2
 ‘They(sg) will not read the book.’
 (12) a-ia βai ne he βaluk na mbuk tea
 NOM-3SG FUT FUT NEG1 read ACC book NEG2
 ‘They(sg) will not read the book.’⁴

⁴ The presence of multiple future tense markers provides a more distal future reading. The two future tense markers *βai* and *ne* appear both alongside one another and separately from one another in our data. Aru (2015) labels *ne* as irrealis rather than future, and does not appear to note any instances of *βai* (which she would represent orthographically as *vai*). The extent to which the two markers are different, and the specific semantics of the future tense reading provided by each one, and by their combination, has been beyond the scope of our investigation.

- (13) a-ia he tuku kaukau tea
 NOM-3SG NEG1 PROG RED.fishing NEG2
 ‘They(sg) don’t go fishing.’

Several generalisations can thus be made from our data so far. First, in a mono-clausal transitive sentence, the two discontinuous negative markers *he* and *tea* encircle the verb and the object. Second, both negation markers are obligatory. Third, this negative marking encircles pre-verbal aspect markers along with the rest of the VP, while pre-verbal tense marking remains outside. Following these generalisations about the structure of Nduindui standard negation, the basic structure of a Nduindui negated sentence can be schematised as in (12):

- (14) Subject Future Tense Neg1 Aspect Verb Object Neg2

The relative order of NEG1 and co-occurring tense and aspect markers schematised above is assumed, as the data gathered in this investigation has found no instances of tense and aspect co-occurring while also negated. Aru (2015) provides evidence of the relative order of future tense *ne* and the aspect marker *tuku* when co-occurring, as below:

- (15) kuru ne tuku lakao
 2DL.SUBJ FUT PROG walk
 ‘You two will be walking.’

(Nduindui; adapted from Aru, 2015, p. 100.)

However, no data in Aru nor our elicitations shows *ne* and *tuku* co-occurring alongside negation. The schema in (14) is thus developed from the relative order of elements in future tense negated clauses and in progressive/continuous negated clauses, as no future tense progressive negated clauses have yet been found. Whether this constitutes simply a gap in the data, or is indicative of complementary distribution resulting from all three being disallowed from co-occurrence requires further investigation. In the following section, we discuss negation in other Vanuatu languages, as a comparative landscape regarding negation in the region.

3. Bipartite Negation in other Vanuatu languages

Nduindui is somewhat unusual within a Vanuatu context by virtue of having only one negation strategy in declarative sentences.⁵ It is common for Vanuatu languages to employ multiple different negation strategies, often conditioned by realis/irrealis, tense and aspect, or sentence type. Many of these languages employ multi-part negation in either one or many of these strategies. This section details the discontinuous negation strategies of a select number of other Vanuatu languages, to illustrate their similarities and differences to the patterns found in Nduindui.

The East-Ambae language Lolovoli, per Hyslop (2001), exhibits a similar, clearly related, strategy of obligatory bipartite negation, involving NEG1 *hi* and NEG2 *tea*:

⁵ The only context in which usage of *he* and *tea* is not the negation strategy is in imperatives: where an alternate (single) pre-verbal negator *hatu* appears.

- (16) go=ni hi hako na hasi tea
 2SG=IRR NEG1 hold ACC bad NEG2
 ‘You will not handle bad things.’

(Lolovoli; Hyslop, 2001, p. 259.)

There is variation in Hyslop’s data surrounding the positioning of NEG2 in relation to full (non-clitic, presumably DP) objects.⁶ Structures with ‘specific’ objects behave the same as the negation patterns of Nduindui, with NEG2 *tea* appearing post-object (see 14), while non-specific subjects appear after NEG2, which thus directly follows the verb, as in (15).

- (17) hi inu na mwetarigelegi tea
 NEG1 drink ACC kava NEG2
 ‘...he didn’t drink the kava.’
- (18) maresu ra=hi inu tea malogu
 child 3NSG.SUBJ=NEG1 drink NEG2 kava
 ‘Children don’t drink kava.’

(Lolovoli; Hyslop, 2001, p. 262.)

A similar pattern is found in Vatlongos, spoken on Ambrym. Per Ridge (2019), the form of NEG1 is conditioned by tense, with *taa* appearing in non-future clauses and *naa* in future. NEG2 *ti* can appear to the right or the left of the object.⁷ In other words, the discontinuous negation appears to sometimes encircle the entire verb phrase, but other times only the verb. Ridge (2019, p. 132 – 135), concludes that this alternation may be conditioned by affectivity of the patient, rather than specificity, as well as the partitive meaning associated with NEG2 *ti*, which has its source in Vatlongos as a partitive marker. In (19), the object of the verb ‘eat’ appears to the left of NEG2, while (20) exemplifies the rightward position of the objects of perception verbs. Complement clauses appear always to the right of NEG2 *ti*. This fact is of particular interest to any analysis attempting to derive the NEG1 V O NEG2 pattern, as will be discussed further in Section 4.

- (19) ra-taa-a hura =ti
 1PL.INCL-NEG1.NFUT-eat yam NEG2
 ‘We don’t eat yam.’
- (20) lu-taa-pus =ti tumen
 3DU.NFUT-NEG1.NFUT-see NEG2 bird
 ‘They didn’t see any birds.’

(Vatlongos; adapted from Ridge, 2019)

Per the descriptions of Ivens (1938) and Walsh (1995), the North-Pentecost language Raga exhibits bipartite negation involving NEG1 *hav* and NEG2 *te(he)*, with some variation in NEG1 form. It would appear, as presented in (19 – 21), that the object consistently appears to the right of NEG2, unlike in Nduindui.

⁶ Per Hyslop (2001), clitic object pronominals are cliticised directly to the verb or adverb. Thus, in the context of negation, Lolovoli NEG2 *tea* always occurs after the entire verb+clitic structure, regardless of specificity of a clitic object.

⁷ Ridge (2019) understands NEG1 *t/naa* to be a prefix, and *ti* to be a clitic. The clitic nature of *ti* may be of particular relevance to its variable positioning.

- (21) yo-n hav van(o) te(he)
 2SG-PST NEG1 go NEG2
 ‘You did not go.’ (Raga; adapted from Walsh, 1995)
- (22) na sav van te dagai nin kimiu
 1SG.SUBJ NEG1 go NEG2
 ‘I will not leave you.’ (Raga; adapted from Ivens, 1938)⁸
- (23) si hav varahu-a te sibo-na
 CONDIT NEG1 save-3SG.OBJ NEG2 self-3SG.POSS
 ‘He cannot save himself.’ (Raga; adapted from Ivens, 1938)

There are, therefore, those related languages in which the post-object NEG2 pattern of Nduindui is not mirrored, and for which any potential analysis may, therefore, need to be markedly different. However, other languages do share the NEG1 V O NEG2 nature of Nduindui, with some data also reflecting a far more definitively clause final NEG2. The Lelepa language spoken on Efate and Lelepa Islands uses NEG1 *ti* and NEG2 *mau*, which Lacrampe (2014) refers to as “particles”. NEG1 appears pre-verbally and NEG2 is positioned clause finally, even when utterances involve adverbials and embedded clauses that appear to be outside the scope of negation:

- (24) e=ti tae takanei e=ga tuagoto mau
 3SG.S=NEG know how 3SG.S=IRR cross NEG2
 ‘He didn’t know how he would cross.’
- (25) se ur=mro ti tae laka mala kasua tapalange mau
 while 3PL.S=again NEG1 able see time hard like.this NEG2
 ‘...and they cannot face hard times like this.’
 (Lelapa; Lacrampe, 2014)

In (21), above, NEG1 appears at the beginning of the matrix clause to the immediate right of a subject clitic. NEG2 appears not at the end of the first clause or verbal complex but following the (non-negative) embedded clause ‘he would cross’, thus appearing completely sentence-finally. In multi-clause constructions like this, when there are more than two clauses the pattern is even more evident. In (23), several listed clauses are present, each one negated. NEG1 appears at the beginning of each clause in the reported speech, while a single NEG2 appears only at the very end of the entire structure, rather than repeated at the end of each individual clause.

- (26) se misi e=lag, “a=ti msau nañit, a=ti
 while missionary 3SG.S=say 1SG.S=NEG1 want mat 1SG.S=NE
 msau wago, a=ti msau nafnag pi kastom
 want pig 1SG.S=NEG1 want food COP custom
 mau
 NEG2
 ‘And the missionary said, “I don’t want mats, I don’t want pigs, I don’t want traditional food.”’
 (Lelepa; Lacrampe, 2014)

⁸ Note that the original examples in (17 – 18) are not glossed in Ivens (1938). We have glossed (18) on the basis of other data, but were able to gloss (17) only partially. Despite the partial gloss, (17) shows that there is material (the object) to the right of NEG2.

The Epi Island languages Lewo (Early, 1994) and Bierebo (Budd, 2010) are well known for their tripartite negation strategies, though this is optional in Lewo and may be interpreted emphatically in Bierebo. In Lewo, NEG1 and NEG2 appear on either side of the verbal complex, while NEG3 is (optionally) positioned further to the right, post-object, even in instances of highly complex objects such as in (27).

- (27) naga ve Ø-va re
 he NEG1.IRR 3SG-go.IRR NEG2
 ‘He will not go.’
- (28) naga pe Ø-pa re poli
 he NEG1.R 3SG-go.R NEG2 NEG3
 ‘He hasn’t gone.’
- (29) naga pe Ø-pisa re suniena tai poli
 he NEG1.R 3SG-say.R NEG2 story ART NEG3
 ‘He didn’t tell a story.’
- (30) pe supē re tai naña kilia suño-ni-la
 NEG1.R chief NEG2 ART REL can go.ahead-TR-3PL
 e yuñæna poli
 PREP work NEG3
 ‘There’s no chief who can lead them in their work.’

(Lewo; adapted from Early, 1994)

Unlike Lelepa, the final negative in Lewo, NEG3 *poli*, appears only ‘within’ the clause relevant to the negative scope, rather than appearing utterance-finally in multi-clause structures like (28).

- (31) teras la nañā pe a-pe praktis re
 youth PL REL NEG1.R 3PL-be practise NEG2
 poli ve a-te re pol e Sarer
 NEG3 NEG1.RR 3PL-kick NEG2 ball PREP Saturday
 ‘The young guys who haven’t practised will not [play] kick ball on Saturday.’

(Lewo; adapted from Early, 1994)

Early (1994) describes some speaker variation regarding optionality of each of the three negative items in Lewo, mostly related to age, and thus presumably a characteristic of language change. This is particularly expected in regard to multi-part negation strategies per Jespersen’s cycle (Jespersen, 1917 & Dahl, 1979). NEG3 *poli* does not occur in irrealis structures in Lewo, as seen previously in (24), nor in negative imperatives. It is also described as able to be replaced by the emphatic particle *nena*, which is translatable as ‘(not) at all’ in these negative contexts. The relationship between negative items and emphasis, particularly on the basis of this pattern of complementary distribution in Lewo, is discussed further in Section 4.2.

4. Nduindui patterns and challenges to existing formal syntactic analyses

4.1. Bipartite negation as a single NegP

Perhaps the most frequently discussed and the most influential formal analysis of bipartite negation comes from Pollock (1989) for French (see also Rowlett 1993 & 1998, Belletti 1990, Ouhalla 1990). It is therefore worth considering if this analysis can be extended to account for the patterns exhibited by the bipartite negation of Nduindui. The core idea in these analyses of French negation is that despite the presence of two discontinuous negative elements, a single NegP is hosting both. Although both Nduindui and French both have bipartite negation in the sense that there are two negation elements present in a single negated clause, the empirical patterns in French are quite different from Nduindui. We therefore argue that attempting to extend such an analysis to Nduindui results in several problems. This is discussed below, starting with the basic premise of this style of analysis of French negation.

An analysis of French negation with one NegP, which is hosting both NEG1 *ne* and NEG2 *pas*, aims to capture the empirical facts of French negation on the basis of a particular structural relationship between the two negators. In Modern (spoken) French, NEG1 *ne* is not obligatory, and is often omitted from the expression of negation in finite clauses.

(32)

- | | | | | | |
|----|--------------------------|-------------|-------------|-------|-------|
| a. | Jean | ne | voit | pas | Marie |
| | John | NEG1 | see.PRS.3SG | NEG2 | Mary |
| | 'John doesn't see Mary.' | | | | |
| b. | Jean | voit | pas | Marie | |
| | John | see.PRS.3SG | NEG2 | Mary | |
| | 'John doesn't see Mary.' | | | | |
| c. | *Jean | ne | voit | Marie | |
| | John | NEG1 | see.PRS.3SG | Mary | |

(French; adapted from Bell, 2004)

The negative *pas* can therefore be used to express negation by itself, as in (30b). In contrast, deletion of *pas*, and the sole use of *ne* to express negation is ungrammatical, as in (30c). In the standard analysis of French negation developed out of Pollock's (1989) work, *ne* is understood as the head of the NegP, while *pas* appears in specifier position. The verb raises to T, and *ne* (a clitic) is argued to move to T as well, attaching to the complex V+T head produced via verb head movement. This gives the surface order *ne V pas*.

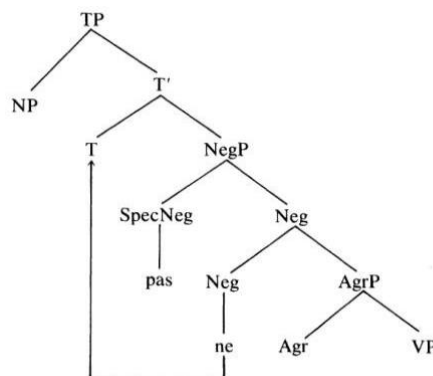


Figure 1: French negation (Pollock, 1989, p. 414.)

Per the NEG Criterion (Haegeman & Zanuttini, 1991 & Haegeman, 1995), the two negation items are in a licensing relationship with one another:

- (33) The NEG Criterion:
- a. A NEG-operator must be in a Spec-head configuration with an X°
 - b. An X° [NEG] must be in a Spec-head configuration with a NEG operator.

In short, this analysis claims Neg $^\circ$ *ne* requires the presence of an (overt) negative operator to express standard verbal negation. This need is satisfied by the negative operator *pas* in the specifier position. However, this operator does not in turn require an overt Neg $^\circ$ to be present, thus allowing *pas* to be used as sole negator in Modern French.⁹ This argument, with parametric variation, has been used in analyses of a variety of other languages with multiple negative markers, such as West Flemish (Haegeman & Zanuttini, 1991), German and Dutch (Haegeman, 1995), varieties of Arabic (Ouhalla, 2002), and at varying times Afrikaans (Haegeman 1995, contra Oosthuizen 1998 & Bell 2004).

We argue in this section that while such a single NegP analysis has been successfully extended to many languages, it is not the right path to understand the negation patterns in languages like Nduindui. One major empirical difference between the bipartite negation present in Nduindui and French is that in Nduindui both negative markers are always obligatory. The foundational issue in attempting to extend a single NegP analysis to Nduindui is immediately clear: unlike in French, there is no empirical evidence in Nduindui that one of the two negative markers is ‘dominant’ in any capacity, can negate by itself without the other, and can therefore be reasonably posited as the negative operator. An analysis with one phrase hosting both negative markers must assume a structural difference between the negative marker hosted in the specifier position and the marker that is the head. An expansion of such an analysis to the extent of the Neg Criterion further embeds this claimed structural relationship between the two negative markers, whereby one requires checking from the other, which is an operator. Again, in the Nduindui data no evidence would support this type of understanding of the language’s negation, as both *he* and *tea* are obligatory. In other words, the two Nduindui markers of negation both resemble French *ne* in that they each require the presence of their counterpart.

Forcing a single NegP analysis also produces difficulties in generating the right word order – more specifically the fact that the object along with the verb is encircled by the two negators. Note that the V→T movement and Neg-raising which happens in French would generate the incorrect order for our Nduindui data: NEG1 V NEG2 O. To derive the NEG1 V O NEG2 order in a single NegP analysis, the following must occur: NEG1 *he* is base generated in Neg $^\circ$, and NEG2 *tea* in the Spec of NegP.¹⁰ Unlike in French, we must posit phrasal movement of the VP into the NegP for the appropriate word order predictions to be made. Therefore, the

⁹ The idea that the Neg $^\circ$ *ne* in French requires an overt operator in its Spec position, but that the NEG operator does not require an overt head counterpart is language specific, not an explicit/universal characteristic of the Neg Criterion (see Haegeman, 1995).

¹⁰ The alternate option, treating NEG1 *he* as positioned in the specifier position and NEG2 *tea* as the head, is not considered here on the basis of two main problems. First, Nduindui exhibits a ban on True Negative Imperatives, whereby *he* + *tea* negation is disallowed in imperative clauses. We assume this follows from a corollary of Zeijlstra’s generalisation (Zeijlstra, 2004, p. 165) regarding the head status of negative markers in languages exhibiting the TNI ban: at least the pre-verbal negator (NEG1 *he*) must be a head in order to explain the presence of the TNI ban in Nduindui. Second, for the VP to raise to SpecNegP and maintain the appropriate surface word order, we must adopt Chomsky’s (1995) proposal that a phrase may have more than one specifier. As with the account discussed in text, the ordering of these specifiers in relation to one another may be a cause for concern.

landing site of this movement must be the specifier, and the Neg^o *he* must further raise out of NegP to derive the appropriate word order. Such an analysis must adopt the presence of multiple specifiers (Chomsky, 1995), as otherwise the specifier position of the NegP is already filled by NEG2 *tea*. If NegP can project more than one specifier in Nduindui, then the VP has a licit landing site. The issue of the ordering of multiple specifiers is one of some contention; this model would require that the externally Merged (i.e., moved) specifier containing the VP would be above the specifier containing NEG2 *tea*, in order to produce the correct surface order. However, this is contra the claims of McGinnis (1998) and Rackowski and Richards (2005) regarding the ordering of specifiers resulting from external and internal Merge. To bypass the stipulation of multiple specifiers, it could be claimed that there is an XP between TP and NegP, and the VP moves to this phrase. This is another unmotivated stipulation, with no motivation for the VP-movement, and also with no principled reasoning for positing an unidentified functional projection except the sole purpose of providing a landing site for the VP. This version of the analysis will also make incorrect word order predictions regarding the position of NEG1 *he*: which must raise higher than the ultimate position of the verb and object, but not higher than the position of future tense. The presence of some stipulative functional projection between NegP and TP will not solve this issue, as NEG1 *he* raising to the head position of TP will produce NEG1 FUT word order. The word order generalisations described in Section 2 regarding the variable positioning of adverbs, and the appearance of embedded clause complements to the left of NEG2 *tea* will also not be captured by this account.

That is, while it is mechanically possible to extend a single NegP analysis to a language like Nduindui, such an attempt results in unmotivated assumptions and movements, and thus has no real explanatory value regarding the nature of Nduindui clausal material and the behaviour of the negative items. On the basis of the Neg Criterion, this analysis directly predicts a type of relationship between an optional NEG1 and NEG2 that is contra the empirical facts of Nduindui. Next, we explore the idea if a better alternative is to discard the single NegP account for obligatory bipartite negation languages, and instead seek an analysis where the two negative items are each housed in separate projections, allowing a model where each is a head.

4.2. Bipartite negation as separate projections: the PolP account

Having argued above that a French-style analysis of bipartite negation is not desirable to extend to Nduindui, in this section we evaluate the potential of a proposal where the Nduindui clause contains both a negation phrase/NegP and a polarity phrase/PolP. In essence, this proposal is an extension of accounts that have been proposed for Negative Concord in Afrikaans (Oosthuizen, 1998; Biberauer, 2007 & 2008). There are two negative markers in Afrikaans, and negation often involves both being used together to express one instance of semantic negation. The two negators bear the same phonological form: *nie*. An example of a negated sentence in Afrikaans is given in (34) below, where the two negators have been represented as *nie1* and *nie2* for ease of exposition. As in Nduindui, direct objects in Afrikaans appear to the left of NEG2. Unlike in Nduindui, NEG1 appears after (tensed) Afrikaans verbs, due to the Verb-Second nature of the language.

- (34) ek ken nie1 dardie man nie2
 1SG know NEG1 that man NEG2
 ‘I don’t know that man.’

(Afrikaans; adapted from Biberauer, 2007, p. 1.)

Following Kandybowicz (2006), Biberauer (2007) claims that the PolP in Afrikaans is associated with an EPP feature which induces clausal pied-piping, allowing an entire CP to move to SpecPolP. This movement is what derives the correct surface order of the negating elements in Afrikaans. A sample derivation is illustrated below for the negated sentence in (35).

- The underlying form of the relevant embedded clause in (33) is given in (36) below, following Oosthuizen's (1998) proposal that *nie1* is head of NegP, while *nie2* is the head of PolP.

- The Pol head has EPP-features that drive the movement of the entire CP to the specifier position of PolP. This generates the surface order of the sentence, as illustrated in (37). The details of extending this style of PolP analysis to Nduindui are discussed below.

- Recall that there are two negators in Nduindui, *he* and *tea*, and that both need to be present to negate a declarative sentence. We propose that NEG2 *tea* can be treated as the head of PolP, and NEG1 *he* as the head of NegP. Taking into account that *he* and *tea* encircle aspect markers, but that the pre-verbal tense markers appear before *he*, we suggest that TP is higher

¹¹ Biberauer (2007) instead claims that *nie1* is adverbial in nature, an AdvP adjoined to the VP (while still the ‘true’ semantic negator). This conclusion is supported in Afrikaans by the ‘optional’ nature of *nie1*: bipartite negation compared to negation via only *nie2* is conditioned in Afrikaans by both clause-type, tense, and semantic scope phenomena. If extending this type of analysis to Ndundui, we argue that an adverbial analysis for either negation marker is not desirable as it leads to the expectation that this adjunct negator will be optional, which is contra the empirical facts detailed in Section 2, where both the negators are shown to be obligatory.

than the NegP and the NegP is higher than AspectP.¹² The PolP is highest in the structure. A proposed full hierarchy of projections is given below:

$$(38) \quad \text{PolP} > \text{CP} > \text{TP} > \text{NegP} > \text{AspP} > \text{vP} > \text{VP}$$

We can retain the idea that there is an EPP feature associated with the PolP that induces clausal pied-piping (Biberauer, 2007), which results in movement of the entire CP to SpecPolP. This is illustrated in the derivation below, for the example previously given in (9), repeated below as (39):

- (39) a-ia ne he βaluk na mbuk tea
 NOM-3SG FUT NEG1 read ACC book NEG2
 ‘They(sg) will not read the book.’

The underlying structure of the relevant constituents is as follows:

$$(40) \quad [\text{PolP } tea \text{ } [\text{CP } [\text{TP } ne \text{ } [\text{NegP } he \text{ } [\text{vP } aia \text{ } [\text{VP } \beta aluk \text{ } na \text{ } mbuk]]]]]]]$$

From this underlying structure in (40), two main movements take place. First, following the vP-internal subject hypothesis (Koopman & Sportiche, 1991), the original position of the subject *aia* is SpecvP. It then moves to SpecTP for the purpose of Nominative Case assignment. Second, the EPP feature associated with PolP drives movement of the entire CP to SpecPolP. These movements generate the surface word order illustrated in Figure 2.

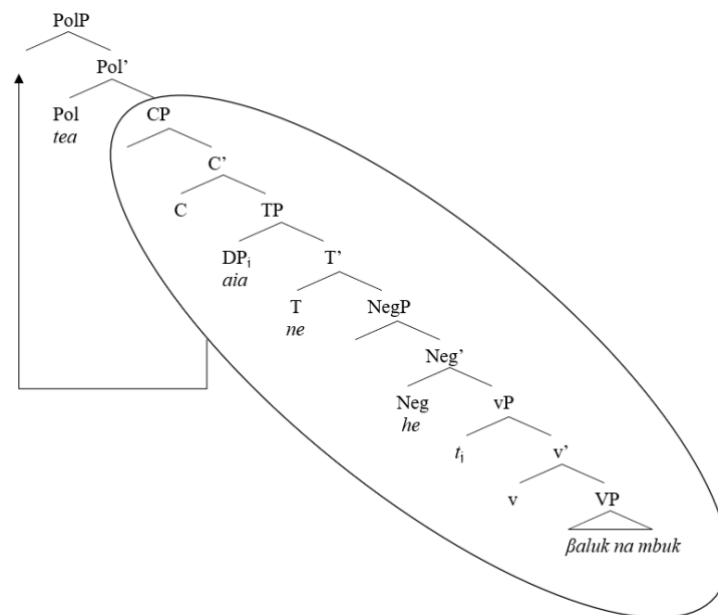


Figure 2: PolP derivation of Nduindui negation

This model could potentially be claimed to be more parsimonious than the single NegP account, as it requires only one movement (albeit clausal pied-piping) in order to produce the correct word order. This model can correctly account for the ordering of aspect marking by positioning

¹² We have not discussed in this paper treating *ne* as head of a MoodP rather than of TP in the interests of space; we have found that doing so cannot ‘save’ any of the approaches evaluated throughout Section 4 given the other word order issues.

AspP lower than NegP. Adoption of a PolP allows us to avoid operator status being conferred onto NEG2 per the Neg Criterion, and instead understand both negative items as heads, which is also a desirable outcome.¹³

However, this analysis also makes strong predictions regarding clause-final material. If the entire CP is being pied-piped to the SpecPolP, there should be no position rightward of the Pol° *tea* for any material to appear in. As seen from the data in Section 2, this prediction does not appear to be borne out. Aru (2015) claims that adverbial material may appear to both the right or to the left of NEG2 *tea* (Aru, 2015, p. 158). Given the derivation sketched out in Figure 2, the word-order predictions of this analysis are found to be incorrect, under the usual assumption that adverbs are adjuncts at the VP-level¹⁴. Bell (2004) notes that modifications to the PolP account could be made to account for these types of patterns, by the addition of a further functional projection between PolP and CP, and that sentence final adverbials could be argued to move prior to the EPP-driven movement of the matrix CP.¹⁵ Further investigation in the Nduindui context would be required to determine if this adaptation would resolve this issue; it is not at this stage strongly empirically motivated in either Afrikaans or Nduindui.

Another argument against the PolP account is the ordering of clausal complements. As described in Section 2, Nduindui embedded clause complements also appear after NEG2 *tea*, rather than the position of standard direct objects. Assuming that clausal complements are generated as complements of the matrix verbs, any movement of the matrix CP upwards into PolP should also take with it the embedded clause. The data thus shows, that NEG2 in Nduindui is not consistently *clause-final*, which a PolP analysis would predict.

Recall, however, the empirical patterns of other Vanuatu languages laid out in Section 3, most notably Lelepa and Lewo. In Lewo, NEG3 *poli* cannot co-occur with the emphatic particle *nená*. If we take the Polarity Phrase to be the position housing emphasis, it may follow that *nená* and *poli* are in complementary distribution because they are each unique Polarity heads and are thus disallowed from co-occurring. The strong clause-final prediction of a PolP analysis appears to be supported in Lelepa on the basis of the ordering of NEG2 *mau*, while Lewo exhibits evidence of an on-going association between NEG3 and emphasis. Both Lelepa and Lewo each therefore exhibit patterns that support the idea that the PolP analysis may be extendable to some Vanuatu languages. Emphasis is a common source of the development of multi-part negation strategies via the Jespersen's cycle (Jespersen, 1917 & Dahl, 1979; see also van der Auwera, 2010 for a more modern overview on the issue of emphasis and the cyclic development of multi-part negation). As such, those languages in which a second negative item (or third in the case of Lewo and Bierebo) is strongly associated with emphasis may still be suited to a PolP based analysis.¹⁶ In sum, the bipartite negation attested in Lolovoli, Lelepa,

¹³ We suggest that neither *he* nor *tea* is a Neg operator, and if there must be a single operator, it be a phonologically null one. The exact syntactic position of this operator and its relation to *he* and *tea*, however, remains outside the scope of this squib.

¹⁴ The only way around it is to postulate that adverbs are adjoined at the PolP level.

¹⁵ Another adaption in the same vein could involve the ordering of CP and PolP. If we adopt a usage of PolP that places it lower than CP, or lower even than TP, a claim could be made that adverbials appearing to the right of NEG2 *tea* are adjoined to CP or TP instead. This would involve making a strong claim about the status of these adverbials, and would need to be tested further in Nduindui. The ordering issue of CP complements, however, would remain an issue.

¹⁶ Per Hyslop (2001), NEG2 *tea* in Lolovoli has the same form as the language's indefinite plural article 'some'. In Nduindui, *tea* is also the form of the numeral meaning 'one'. Clark (1985, p. 209) reconstructs a pre-verbal Proto-North-Central-Vanuatu negative marker **(st)a(vb)V* and a post-verbal **tea*, and presumes the post-verbal marker to have a relation to PNCV **tea* 'one'. He posits that NEG2 **tea* may have developed from emphatic use of numeral **tea* in a negative context ("not one!" p. 209). We have found at this stage no synchronic evidence of speakers using the numeral *tea* in Nduindui as an emphasis strategy. On the basis of reconstruction and related

and Lewo all suggest that a PolP analysis of negation can be extended to these languages, capturing both the emphasis patterns as well as clause-final position of NEG2. However, in Nduindui, we conclude that the empirical evidence cannot support an analysis that adopts Oosthuizen's (1998) and Biberauer's (2007, 2008) style of PolP.

4.3. Bipartite negation as separate projections: NegP inside a NegP

Bell (2004) claims that the Afrikaans evidence does not support the existence of a PolP in the language, and taking adverbial and movement data as empirical support, claims that *nie1* and *nie2* cannot be housed in two entirely separate XPs. He proposes instead an “expanded” (or articulated) NegP containing two heads. This proposal involves a NegP-inside-a-NegP type structure, as below in Figure 3.

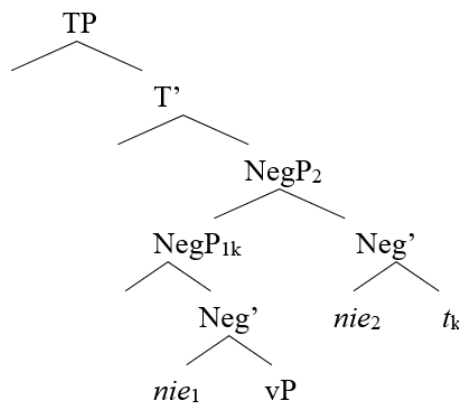


Figure 3: Afrikaans “expanded NegP” (adapted from Bell, 2004)

NegP₂ dominates NegP₁, the phrases each containing heads *nie2* and *nie1* respectively. The entirety of NegP₁ is pied-piped along with *nie1* when it moves into SpecNegP₂ in order to satisfy a checking/agreement relationship between *nie1* and *nie2*.¹⁷ This pied-piping thus includes the vP and VP dominated by NegP₁, which in Afrikaans results in the appropriate *nie1* O *nie2* word order because the verb raises further out of the VP. If we assume that in Nduindui the verb does not raise out from its base position as head of VP, we can thus derive NEG1 V O NEG2 word order in the same manner. A basic tree derivation is sketched out below in Figure 4.

language data, however, it seems reasonable to assume that the development of NEG1 *tea* from numeral *tea* was at some point generally related to emphatic negation, although the material is not primarily an intensifier outside of this context.

¹⁷ Bell (2004) specifically claims that *nie1* contains an interpretable negative feature [*i*NEG], while *nie2* has an uninterpretable [*u*NEG] feature, which thus drives internal Merge (movement) of the phrase containing *nie1* to the specifier position of NegP₂, for the purposes of valuing the uninterpretable feature against its interpretable counterpart prior to Spell-Out.

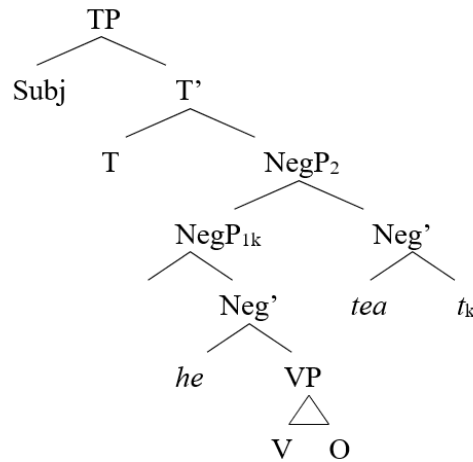


Figure 4: Expanded NegP for Nduindui

However, this account faces the same challenges as a PolP account because of the nature of the movements involved. Pied-piping the entire constituent containing the vP and VP up ‘past’ the head position filled by NEG2 *tea* means that this analysis makes the incorrect predictions regarding both the ordering of adverbs and CP complements post-NEG2. As with the PolP analysis, there is no position to the right of NEG2 *tea* available for any material, and adjuncts within the vP will always appear to the left of NEG2 *tea*. Given Aru’s (2015) claim that adverbs appear both before and after NEG2 *tea*, it could be worthwhile to investigate if any reasonable claim could be made regarding the status of those adverbs which appear in Adv NEG2 order compared to those in NEG2 Adv order. If there is a pattern regarding certain adverbs only appearing to the left of NEG2, while others appear to the right, perhaps a claim could be made regarding TP or CP adjunction of certain adverbs. Any type of claim regarding this would need to be motivated by a clear empirical pattern, which is not available at this stage. The CP complement ordering would, however, remain a concern as it has with the other analyses.

We conclude that in Nduindui, although both the PolP analysis and Bell’s (2004) expanded NegP analysis thus may appear more parsimonious than any attempt to extend a single NegP analysis following Pollock (1989), both approaches cannot appropriately account for the empirical facts regarding the ordering of adverbs and complement clauses.

4.4. Bipartite negation as separate projections: two NegPs and VP-raising vs V-raising

The expanded NegP account of Bell (2004) does, however, exemplify a type of analysis that may be worthwhile to explore in languages exhibiting bipartite negation. Simply proposing the presence of multiple Negative Phrases, with NEG1 and NEG2 each housed in a separate NegP, allows us to retain the status of both NEG1 and NEG2 as heads, without resorting to the presence of a unique PolP. The expanded NegP ‘version’ of two NegPs involves an Agree relationship between *nie1* and *nie2* motivating movement of one NegP into the specifier position of the other. There is another possibility that is worth evaluating: an analysis that still involves two separated NegPs present in the clausal structure, but which does not involve Neg-motivated movement, and where one NegP is not nested under the other. We spell out such an analysis below.

Recall the generalised Nduindui sentence structure presented in Section 2, repeated below.

(41) Subject Future Tense Neg1 Aspect Verb Object Neg2

If we take the ordering of both the verb and object in relation to NEG1 and NEG2 as symptomatic of VP-raising past the position of NEG2, we can propose a hierarchy of projections that positions the two NegPs in such a way that the surface word order is straightforwardly accounted for in the basic clause structure. Let us assume that NEG1 *he* is the head of a higher NegP₁, and that NEG2 *tea* is head of the lower NegP₂. As tempting as it may be to propose a hierarchy of projections where NegP₁ sits between TP and AspP, and NegP₂ appears below AspP, we find that phrasal raising of the VP to the specifier position of a NegP₂ can never appropriately deal with the Nduindui facts. As has been the problem with the type of phrasal movements associated with both the PolP and Expanded NegP analyses, any phrasal movement of the entire VP constituent higher than vP, to the left of the position of NEG2 *tea* will predict the incorrect word order of both adverbs adjoined within the vP and clausal complements. Although we can derive the correct word order of verb and object via VP-movement, this VP movement will always predict the raising of non-object complements of the V head as well. As illustrated below in Figure 5 this derivation runs in trouble with the adverb orders; as NegP₂ dominates vP, this analysis would predict adverbs would never intervene between the object and NEG2 *tea*, contra empirical facts (the ADV Neg2 order, Aru 2015, p. 159).

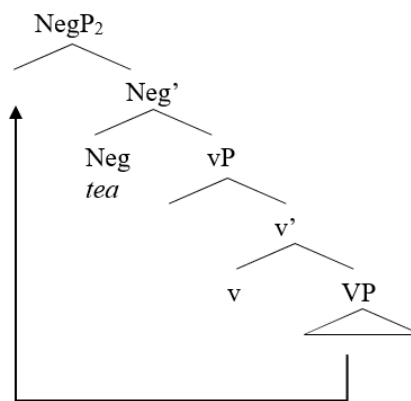


Figure 5: VP movement to SpecNegP₂

Note that the movement of the VP is also stipulative in nature; implying an entirely distinct clausal derivation in the presence of negation than in standard declarative clauses.¹⁸ Unlike in the case of the PolP or Expanded-NegP accounts, this style of movement cannot be explained via Agreement/checking requirements between the two negative heads, as the constituent being moved is the VP. Thus, despite the technical parsimony associated with a derivation involving only one movement, this movement is poorly motivated. As with all the analyses evaluated in this section so far, this account also struggles with the word order facts associated with adverb phrases and complement clauses. These two issues thus remain consistently of concern in positing any reasonable derivation.

The ordering of clausal complements to the right of NEG2 leads to difficulties when attempting to utilise VP raising as a part of the standard derivation of the Nduindui clause. If the VP raises, the complement within it should also. One possibility to explain the difference in behaviour between standard direct objects and embedded clausal complements could be the

¹⁸ Any more standard instance of VP-raising into the Tense or Inflectional domain of the clause struggles to account for the basic word order facts regarding the order of subject, tense and aspect markers in relation to the verb and object constituent.

Case requirements of nominal objects and associated movement, coupled with a V-raising account (instead of a VP-raising approach). Adopting a Agr(O)P account of accusative case assignment (Chomsky, 1991) means that nominal DOs must exit the VP for the purposes of Case feature checking, while any CP complement will remain in the VP. This could explain why direct objects appear to the left of NEG2, while CP complements appear to the right: the nominal DOs are moving above NegP₂, while CPs are not. If it is the verb *head* which raises, rather than the entire VP, this could explain the word order alternation. However, given the order of the other elements in the clause, attempting to utilise head movement of the verb will predict incorrect word orders under a strict Head Movement Constraint (Travis, 1984; hereafter HMC). We have established that the ultimate position of the verb and object at Spell-Out must be higher than the position of NEG2 *tea*; thus, if the V head is moving separately from the object, it must be moving to a phrase higher than NegP₂, to allow for the object to intervene between the two. However, given the HMC, any head movement of the verb must move first to the head position of NegP₂. Given that this node is filled by the Neg head *tea*, head movement through this position will “roll up” the Neg head, creating now a complex head V+*tea*. Regardless of where this complex head moves to, the object should now not be able to intervene between the verb and NEG2 *tea*. To make this analysis work, it has to include an additional, independent component that prevents such interference effects, e.g. “long head movement” which does not obey the HMC (Paul & Potsdam, 2024).¹⁹

That is, while V-raising is perhaps as stipulatory as VP-raising in the Nduindui context, a V-raising can offer a technical solution to account for the clausal complement order, which has remained a problem for all the other analyses discussed in this paper. Such an analysis can also have important consequences for some other languages of Vanuatu, e.g. Vatlongos. Recall that in Vatlongos, NEG2 *ti* can appear to the right or the left of the object (Ridge 2019). It can potentially be argued that there are some objects that need to move out of the VP to get Case, while some objects do not²⁰. When objects move out of the VP, they land in a position higher than Neg2, creating an order where the object is to the left of Neg2. Objects which do not undergo such movement remain to the right of Neg2. Future work must focus on exploring empirical evidence that can potentially support this style of analysis for Nduindui as well as for some other languages like Vatlongos.

5. Conclusion

In this paper, we have presented novel patterns regarding clausal negation in Nduindui leading to several empirical generalisations. We have demonstrated that (a) both the negation elements *he* and *tea* are obligatory in negating a clause, (b) these elements obligatorily encircle the verb and object, and (c) aspect markers, when present, are also encircled by the negators, while tense markers remain to the left of NEG1. To capture these generalisations, we first argued that a Pollock (1989) style analysis cannot be extended to Nduindui, and then presented evaluations

¹⁹ Another potential explanation may be that the difference between DP objects and clausal complements is one of weight; that is, complexity. The tendency for “heavy” constituents to appear toward the end of the utterances has been well-established cross-linguistically, such as the availability of Heavy NP Shift in English (see Ross, 1967). Perhaps a pied-piping explanation could claim that that extremely heavy constituents are stranded, rather than pied-piped in this context in Nduindui. This could perhaps ‘save’ the use of VP raising, but requires additional stipulation that need to be investigated further.

²⁰ One well-known example is Niuean, where DP-objects move out, but NP-objects do not (Massam 2001). It is possible that some characteristics of the noun might determine if it needs to move out of the VP for Case in some languages of Vanuatu, e.g. animacy, definiteness, etc.

of three alternate accounts: the PolP account of Oosthuizen (1998; see also Biberauer, 2007 & 2008), the Expanded/Articulated NegP of Bell (2004), and the possibility of two separated NegPs. We argue that obligatory multi-part negation languages are best suited to analyses where each negator is a head, unlike a single NegP account such as Pollock's. However, attempting to fully extend the PolP or expanded NegP analyses to Nduindui faces drawbacks because of the strong word order predictions that the large-scale phrasal pied-piping in each of these accounts produces. We have discussed how abandoning each of these accounts in favour of two entirely separate NegPs has its own problems, but we have suggested that such an account, if coupled with V-raising as well as Case-related movement of the object-DP, can account for the patterns related to embedded complement clauses. The multi-part negation strategies of other Vanuatu languages have also been discussed in Section 3, and on the basis of the attested patterns of several of these languages, we have suggested that the PolP account may have potential as a future avenue for structural accounts of obligatory discontinuous negation in some Vanuatu languages (e.g. Lelepa and Lewo), despite our conclusion that it is not plausible in an Nduindui context.

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